

District Dataset and Data Dictionary Report

Updated Edition — Every Column Explained in Plain Language

Internship Project — Minorities Early-Warning System

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Contents

A note on the placeholders in this document	2
Plain-Language Summary First	2
1. The Working Table — <code>minority_incidents</code>	2
1.1 Where it lives	2
1.2 What’s in it	3
1.3 Quick numbers	3
2. Column-by-Column Walkthrough	3
2.1 Identifier columns (3)	3
<code>id</code> — Internal database ID	3
<code><case_id></code> — Official PSCA case reference	4
<code><dispatch_id></code> — PSCA system lead ID	4
2.2 Time columns (4)	4
<code>incident_date</code> — The day the case was reported	4
<code><received_time></code> — Exact timestamp the call was accepted	4
<code>incident_year</code> , <code>incident_month</code> — Numeric breakouts	5
2.3 Location columns (7)	5
<code>district_id</code> — Numeric district identifier	5
<code>district_name</code> — Human-readable district name	5
<code>police_station_id</code> — Specific police station code	5
<code>police_station</code> — Human-readable PS name	6
<code>lat</code> , <code>long</code> — GPS coordinates of the incident	6
<code>is_lahore</code> — Quick boolean: is this row from Lahore?	6
2.4 Case-nature taxonomy (3)	6
<code><category_lvl1></code> — Top-level category	6
<code><category_lvl2></code> — Mid-level category	7
<code><category_lvl3></code> — Most specific category	7
2.5 Free text + derived signals (5)	7
<code>description</code> — Caller’s verbatim words	7
<code>match_source</code> — How did this row qualify?	8
<code>matched_keywords</code> — Which words triggered the keyword filter	8
<code>minority_community</code> — Inferred community	8
<code>is_minority_targeted</code> — The STRICT label	9
2.6 Metadata (1)	9

loaded_at — When this row was loaded	9
3. The District-Level Severity Dataset	9
3.1 What's in it	9
3.2 How severity is assigned	9
3.3 What it shows	10
3.4 Caveat about the severity bands	10
4. Six More Tables Derived From This One	10
5. Two More Tables For Predictions	11
6. Known Data-Quality Issues	11
7. How to Refresh This Table	11
8. How to Query It Yourself	12
9. References	12

A note on the placeholders in this document

This is the **public release** of the data dictionary. Internal PSCA schema details (source database names, exact column names of the Emergency-15 case log) have been replaced with generic placeholders in angle brackets — for example `<source_call_log_table>`, `<category_lv12>`, `<case_id>`. The methodology, structure, and field meanings are unchanged; only the institutional names have been genericized so the document is safe to share publicly.

If you receive the full project bundle as a trusted recipient, the internal data dictionary in that bundle uses the real names.

Plain-Language Summary First

If you're new to this kind of document, here's what it is:

A **data dictionary** is just a list of every column in a table, what's in it, what it means, and where it came from. It's like the table of contents at the front of a book — you can look up anything without reading the whole book.

This document is the data dictionary for the working table at the heart of the Minorities Early-Warning System. The table is called `minority_incidents`, and it has **4,110 rows** (one row per phone call to PSCA's Emergency-15 helpline involving a religious minority) and **23 columns** (different pieces of information about each call).

The document also explains the **district-level severity dataset** — a summary CSV that aggregates the 4,110 calls up to the district level and assigns each district a severity rating (Low / Medium / High / Critical).

1. The Working Table — `minority_incidents`

1.1 Where it lives

Field	Value
Database	<analytics_db> (PostgreSQL)
Table name	minority_incidents
Rows	4,110
Columns	23
Built from	<source_call_log_table> (a 5.7-million-row source)
Build script	code_reference/data/build_minority_incidents.py

1.2 What’s in it

Each row is one phone call to Emergency-15 where the call satisfies **either** of these two filters:

1. The operator tagged the call as a “**Religious Offence**” in the case-nature hierarchy, **OR**
2. The caller’s description **mentions a minority community by name** (Christian, Ahmadi, Hindu, Sikh) or a place of worship (church, temple, gurdwara) or a religious-accusation keyword (blasphemy).

This is a deliberately wide filter — we’d rather pick up too many cases than miss real ones. A stricter label (`is_minority_targeted`) is provided as a column so downstream code can filter to only the high-confidence positives.

1.3 Quick numbers

Slice	Count	% of total
All rows	4,110	100%
Rows from Lahore	928	22.6%
Strict-label minority-targeted	312	7.6%
With valid GPS coordinates	3,281	79.8%
Year 2024	1,450	35%
Year 2025	2,117	52%
Year 2026 (Jan–June)	525	13%

2. Column-by-Column Walkthrough

There are 23 columns. We group them into 6 logical buckets to make it easier to follow.

2.1 Identifier columns (3)

These uniquely identify each row in the table.

id — Internal database ID

Property	Value
Type	Integer (auto-increment)
Example	1234
What it means	A unique number assigned to each row when it was loaded. Has no meaning outside this table — it’s just a row-counter.

Property	Value
Why it's here	So that other tables (predictions, features) can link back to a specific row.

<case_id> — Official PSCA case reference

Property	Value
Type	Text
Example	LHR-20260615-323147436
What it means	The official case identifier assigned by Emergency-15 when the call was logged. Same format as PSCA uses elsewhere.
Why it's here	So a human analyst can look up the same case in PSCA's other systems.
Privacy note	This is internal PSCA reference data. Do not share publicly.

<dispatch_id> — PSCA system lead ID

Property	Value
Type	Text
Example	15612345
What it means	Another PSCA-internal identifier, used for cross-referencing with the lead-management system.

2.2 Time columns (4)

When did the incident happen?

incident_date — The day the case was reported

Property	Value
Type	Date (YYYY-MM-DD)
Example	2026-06-15
What it means	The calendar date the call was placed.
Source	<source_call_log_table>.date (parsed from text)

<received_time> — Exact timestamp the call was accepted

Property	Value
Type	Timestamp (YYYY-MM-DD HH:MM:SS)
Example	2026-06-15 14:23:08

Property	Value
What it means	The exact second the call was picked up by an operator.
Used for	Computing “prior 30 days” counts — we use this for strict temporal ordering.

incident_year, incident_month — Numeric breakouts

Property	Value
Type	Integer
Example	2026, 6
What it means	Year (4 digits) and month (1–12) extracted from <code>incident_date</code> . Convenience columns to make grouping by year/month easy.

2.3 Location columns (7)

Where did the incident happen?

district_id — Numeric district identifier

Property	Value
Type	Text (numeric string)
Example	40 (Lahore), 25 (Multan)
What it means	The Emergency-15 system’s district identifier. The same system is used across PSCA, the police boundaries database, and Emergency-15 dispatch.
Lookup	Use <code><station_boundaries_ref>.district_id</code> to resolve.

district_name — Human-readable district name

Property	Value
Type	Text
Example	Lahore, Multan, Faisalabad
Source	Resolved from <code>district_id</code> via the <code><station_boundaries_ref></code> table

police_station_id — Specific police station code

Property	Value
Type	Text (numeric string)
Example	657

Property	Value
What it means	The Emergency-15 identifier for the specific police station responsible for this geographic area. Punjab has 656 such police stations.

police_station — Human-readable PS name

Property	Value
Type	Text
Example	Mughalpura, Old Anarkali, Garhi Shahu

lat, long — GPS coordinates of the incident

Property	Value
Type	Floating-point number
Example	lat: 31.5615, long: 74.3095
What it means	Latitude and longitude where the caller's phone said they were. Some are exact incident locations; some are police-station centroids (when the caller location couldn't be determined).
Coverage	About 80% of rows have valid coordinates. The rest have NULL for these fields.
Privacy note	These are precise to ~3 decimal places, about 100 meters. Anyone with the lat/long can pinpoint a house. For external presentation, consider rounding to 3 decimals (~110m) to reduce precision.

is_lahore — Quick boolean: is this row from Lahore?

Property	Value
Type	Boolean (TRUE / FALSE)
Derived from	district_id = '40'
Why it exists	Lahore is the focal city of the project. Having a boolean makes filtering trivially fast.

2.4 Case-nature taxonomy (3)

What kind of case is this?

PSCA's case-nature hierarchy has 3 levels — broad to specific.

<category_lv11> — Top-level category

Property	Value
Type	Text
Examples	Crime Against Person, Crime Against Property, Law & Order, Other Help
Notes	A handful of high-level buckets, ~12 distinct values.

<category_lv12> — Mid-level category

Property	Value
Type	Text
Most relevant example	Religious Offences
Other examples	Theft, Robbery/Snatching, Assault, Public Disorder
Notes	Roughly 40 distinct values. Religious Offences is the critical one for this project.

<category_lv13> — Most specific category

Property	Value
Type	Text
Examples (under Religious Offences)	Any Other Religious Issue, Defiling of Holy Book, Derogation of Holy Persons, Distribution / Display of hateful sectarian material, Hate Speech, Attack/Damage of Religious Places (Church/Mosque/Gurdwara/Mundar/Imambargah/Mazzar)
Notes	~150 distinct values. The 6 listed above are all the Level-3 sub-categories of “Religious Offences”.

2.5 Free text + derived signals (5)

The caller’s verbatim words and the things we extracted from them.

description — Caller’s verbatim words

Property	Value
Type	Text (up to 2000 characters)
Example	"caller reported some persons broke the Christmas decorations need help"
What it is	The operator’s verbatim transcription of what the caller said. Often a mix of English, Urdu (transliterated), and operational shorthand.

Property	Value
Privacy warning	Contains phone numbers, addresses, individual names, CNIC numbers, and call-center metadata. It is auto-redacted before reaching any dashboard, but the raw column in this table is the original unredacted text . Do not share publicly.
What's done with it	The system extracts community keywords from this field to derive <code>minority_community</code> . The dashboard displays a redacted preview (and only when explicitly toggled on).

`match_source` — How did this row qualify?

Property	Value
Type	Text (enum)
Possible values	<code>religious_offence</code> , <code>desc_keyword</code> , <code>both</code>
What it means	Tells you <i>which</i> of the two filter clauses this row satisfied. <code>religious_offence</code> = formally tagged. <code>desc_keyword</code> = caught only by description text. <code>both</code> = both — the highest-confidence subset.
Distribution	<code>religious_offence</code> : 2,463 (60%) · <code>desc_keyword</code> : 1,312 (32%) · <code>both</code> : 335 (8%)

`matched_keywords` — Which words triggered the keyword filter

Property	Value
Type	Text array
Example	<code>{christian, blasphemy}</code>
What it means	Every minority-community or blasphemy-related word the regex found in the description.

`minority_community` — Inferred community

Property	Value
Type	Text (enum)
Possible values	<code>christian</code> , <code>ahmadi</code> , <code>hindu</code> , <code>sikh</code> , <code>multiple</code> , <code>unspecified</code>
Distribution	<code>unspecified</code> : 2,530 · <code>christian</code> : 1,305 · <code>ahmadi</code> : 166 · <code>hindu</code> : 58 · <code>sikh</code> : 47 · <code>multiple</code> : 4
How it's inferred	A case-insensitive regex over the description. <code>christian</code> matches “christian”, “christians”, “church”, “pastor”, “masihi”. <code>ahmadi</code> matches “ahmadi”, “qadiani”, “ahmedi”. And so on.

Property	Value
unspecified meaning	The description doesn't name a community. It doesn't mean the case is <i>not</i> minority-related — it might just not say the community out loud.

`is_minority_targeted` — The STRICT label

Property	Value
Type	Boolean (TRUE / FALSE)
Definition	<category_lvl2> = 'Religious Offences' AND minority_community IN ('christian', 'ahmadi', 'hindu', 'sikh', 'multiple')
Distribution	TRUE: 312 (7.6%) · FALSE: 3,798 (92.4%)
Why it matters	This is the target the prediction models learn from. It's the high-confidence “yes, this is minority-targeted” subset.

2.6 Metadata (1)

`loaded_at` — When this row was loaded

Property	Value
Type	Timestamp
Example	2026-06-22 12:33:45.428463
What it means	When the build script populated this row. Useful for audit (“when did we last refresh?”).

3. The District-Level Severity Dataset

Alongside the case-level `minority_incidents` table, the system produces a smaller summary CSV: <see `project_intermediate_data`>. This aggregates all 4,110 cases up to the district level and assigns each district a severity rating.

3.1 What’s in it

Column	Type	Example
District	Text	Lahore, Faisalabad
Total Cases	Integer	928, 381
Severity_level	Text	Critical, High, Medium, Low
Reporting Period	Text	Current Dataset (2024 → 2026 YTD)

3.2 How severity is assigned

Cases	Severity
0 – 50	Low
51 – 150	Medium
151 – 300	High
300 +	Critical

These bands come from the original Research Design Report. We use them unchanged for compatibility.

3.3 What it shows

District	Cases	Severity
Lahore	928	Critical
Faisalabad	381	Critical
Sheikhupura	282	High
Sialkot	258	High
Gujranwala	233	High
Bahawalpur, Multan, Rahim Yar Khan...	50–150	Medium
Smaller districts	< 50	Low

3.4 Caveat about the severity bands

These bands are **count-based**, not intensity-based. A district with 194 minor public-disorder calls is rated the same severity as a district with 194 mosque attacks — even though those are operationally very different.

The policy brief explicitly flags this as a recommendation for the next version: split “volume severity” from “intensity severity”.

4. Six More Tables Derived From This One

The case-level `minority_incidents` table is the foundation. Six feature tables, one per predictor family, are derived from it (and from external calendars):

Table	Rows	Cols	What it adds
<code>minority_features_proximity</code>	density	11	Rolling case counts at PS and district level
<code>minority_features_religious_calendar</code>	religious	12	Days to / since major religious events
<code>minority_features_political_calendar</code>	political	7	Political event
<code>minority_features_misinformation</code>	misinformation	7	proximity Misinformation event flags
<code>minority_features_responsiveness</code>	responsiveness	6	Local PS response-time baseline
<code>minority_features_sentiment</code>	sentiment	6	Placeholder (all NULL until a feed is connected)

Table	Rows	Cols	What it adds
minority_features_merged	4,110	62	All six joined together — what the models train on

Each feature table has its own schema file (`code_reference/features/schema_*.sql`) and its own builder script (`code_reference/features/*.py`).

5. Two More Tables For Predictions

Table	Rows	What it stores
minority_predictions	4,110	LR and RF model scores for every case in minority_incidents
minority_pswweek_train	82,000	Training data for the PS-week forecaster: one row per (police station $\times$ week) for 2024-2026
minority_pswweek_forward	656	Current forward prediction for every PS — risk of an incident in the next 30 days

6. Known Data-Quality Issues

We catalog these honestly:

1. **The descriptions are operational, not analytical.** They mix English, Urdu, Roman Urdu, and operator shorthand. Spelling is inconsistent. The keyword extractor catches the obvious patterns but will miss creative spellings.
2. **“Unspecified” doesn’t equal “not minority-related”.** Many cases have a generic description without a community name — the operator was busy, the caller didn’t say, or they used “they/those people” instead of naming a community. These cases are still in the table.
3. **About 20% of rows have NULL or invalid lat/long.** These are kept in the count tables but excluded from the map.
4. **Some older lat/longs are police-station centroids, not incident locations.** This is a known upstream issue in `<source_call_log_table>`. Recent rows (2025-onwards) have better GPS data.
5. **24 rows had coordinates outside Pakistan entirely** (data-entry errors — swapped lat/long, zero coords, etc.). These are filtered out at the dashboard layer.
6. **Reporting bias is irreducible.** A community that under-reports will be under-represented in this dataset. We say this loudly throughout the documentation.

7. How to Refresh This Table

To re-pull the latest data from the source `<source_call_log_table>` table:

```
cd minorities_project
python3 code_reference/data/build_minority_incidents.py
```

The script drops and recreates the table from scratch. It is **idempotent** — running it twice in a row produces the same table.

To export the latest version to CSV (for the portable bundle):

```
python3 code_reference/data/export_snapshot.py
```

8. How to Query It Yourself

In psql:

```
\c db_predictive_policing
```

```
-- All Lahore strict-targeted cases this year
SELECT <case_id>, incident_date, police_station, minority_community
FROM minority_incidents
WHERE is_lahore = TRUE
      AND is_minority_targeted = TRUE
      AND incident_year = 2026
ORDER BY incident_date DESC;
```

```
-- Top 10 police stations by case count
SELECT police_station, COUNT(*) AS n
FROM minority_incidents
WHERE is_lahore
GROUP BY police_station
ORDER BY n DESC
LIMIT 10;
```

```
-- Community breakdown for the strict-label set
SELECT minority_community, COUNT(*)
FROM minority_incidents
WHERE is_minority_targeted
GROUP BY minority_community;
```

In Python:

```
import pandas as pd
df = pd.read_csv('<see project CSV exports>',
                parse_dates=['incident_date'])
df_lahore = df[df['is_lahore'] == True]
print(df_lahore['minority_community'].value_counts())
```

9. References

- Builder script: code_reference/data/build_minority_incidents.py
- DDL (table schema): code_reference/data/schema.sql
- CSV snapshot: <see project CSV exports>
- Source table: <source_call_log_table>

- **Boundaries (district / police-station resolution):** <analytics_db>.<station_boundaries_ref>

This is the updated edition of the original PSCA “District Dataset and Data Dictionary Report”, reflecting the case-level working table that powers the Minorities Early-Warning System.